Computational Modeling - Assignment 1

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In this assignment we learn how to assess rates from a binomial distribution, using the case of assessing your teachers' knowledge of CogSci

N.B. this markdown has 2 parts as it spans 2 weeks of teaching

First part

You want to assess your teachers' knowledge of cognitive science. "These guys are a bunch of drama(turgist) queens, mindless philosophers, chattering communication people and Russian spies. Do they really know CogSci?", you think.

To keep things simple (your teachers should not be faced with too complicated things): - You created a pool of equally challenging questions on CogSci - Each question can be answered correctly or not (we don't allow partially correct answers, to make our life simpler). - Knowledge of CogSci can be measured on a scale from 0 (negative knowledge, all answers wrong) through 0.5 (random chance) to 1 (awesome CogSci superpowers)

This is the data: - Riccardo: 3 correct answers out of 6 questions - Kristian: 2 correct answers out of 2 questions (then he gets bored) - Daina: 160 correct answers out of 198 questions (Daina never gets bored) - Mikkel: 66 correct answers out of 132 questions

Questions:

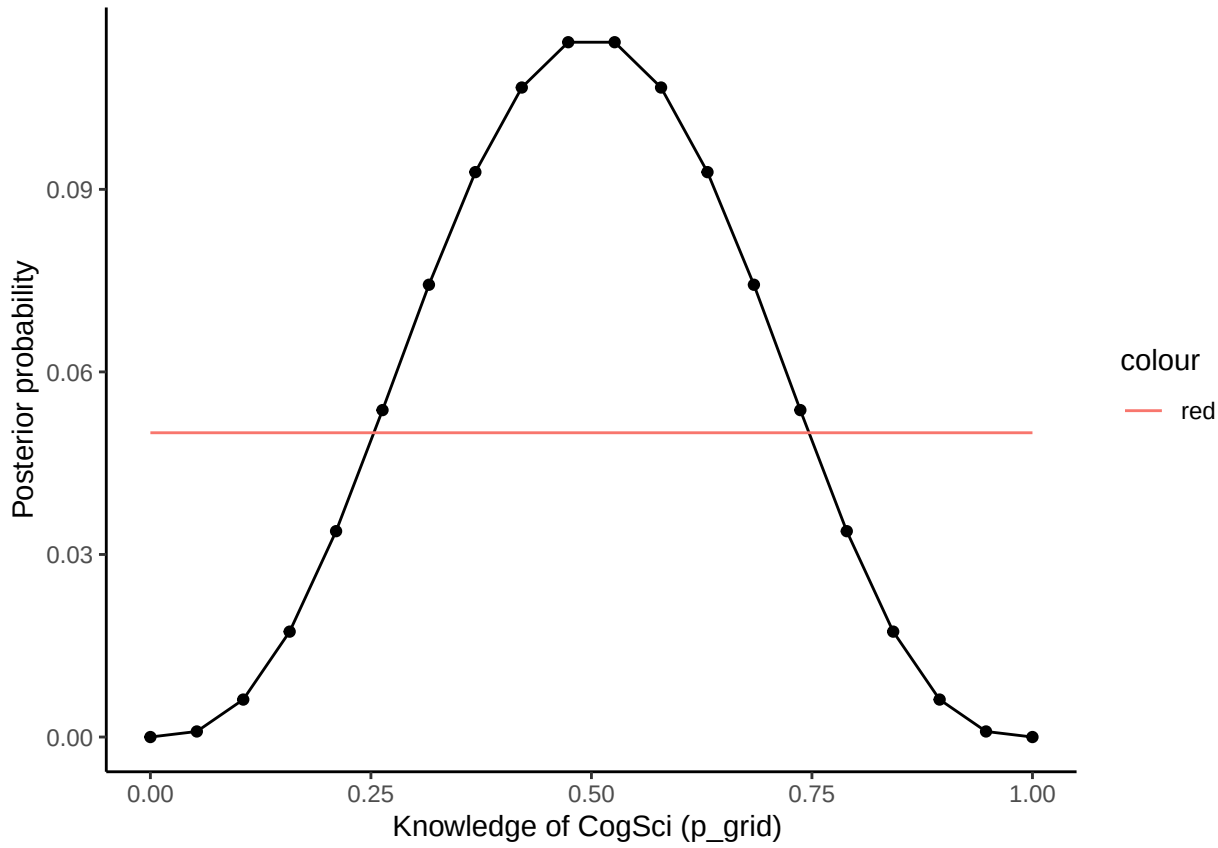
```
############################ LOADING PACKAGES ################################
pacman::p_load(tidyverse, rethinking, patchwork)
```

1. What's Riccardo's estimated knowledge of CogSci? What is the probability he knows more than chance (0.5) [try figuring this out. if you can't peek into chapters 3.1 and 3.2 and/or the slides]?

- First implement a grid approximation (hint check paragraph 2.4.1!) with a uniform prior, calculate the posterior and plot the results

```
############################ THE RICCARDO KNOWLEDGE ################################
# Number of grid points
dens = 20

# Making grid approximation and uniform prior
riccardo <- tibble(
  p_grid = seq(0,1, length.out = dens), # Defining grid
  prior = dunif(1) # Making a uniform prior
) %>%
# Calculating the likelihood, unstd. posterior and std. posterior
```

```r
  mutate(
    likelihood = dbinom(3, size = 6, prob = p_grid),
    unstd_post = likelihood*prior, # unstd. posterior
    posterior = unstd_post/sum(unstd_post) # std. posterior
  )

# PLOTTING
ggplot(riccardo, aes(p_grid, posterior)) +
  geom_point() + geom_line() +
  geom_line(aes(p_grid, prior/dens, color = 'red')) +
  xlab('Knowledge of CogSci (p_grid) ') +
  ylab('Posterior probability') +
  theme_classic()
```




```r
# The probability that Riccardo knows more than chance
mean(riccardo$p_grid > 0.5)
```

```
## [1] 0.5
```

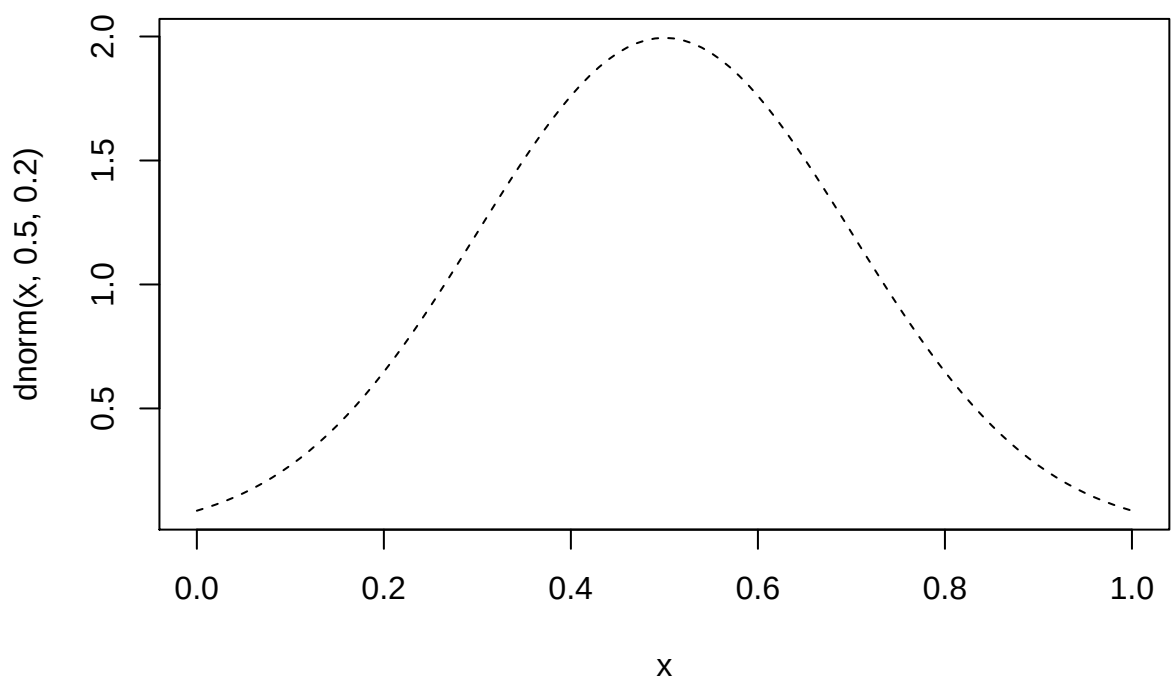
```r
# There is a 50 % probability that Riccardo knows more than chance.
```

- Then implement a quadratic approximation (hint check paragraph 2.4.2!).

```r
############################ RICCARDO WITH QUADRATIC APPROXIMATION ##############################
# Quadratic approximation
qa <- quap(
  alist(
    correct ~ dbinom(correct + false, p), # likelihood
    p ~ dunif(0,1) # uniform prior
  ),
  data = list(correct = 3, false = 3)
)

precis(qa) # mean = 0.5, sd = 0.2
```

```
##        mean        sd      5.5%     94.5%
## p 0.4999992 0.2041233 0.1737707 0.8262277
```

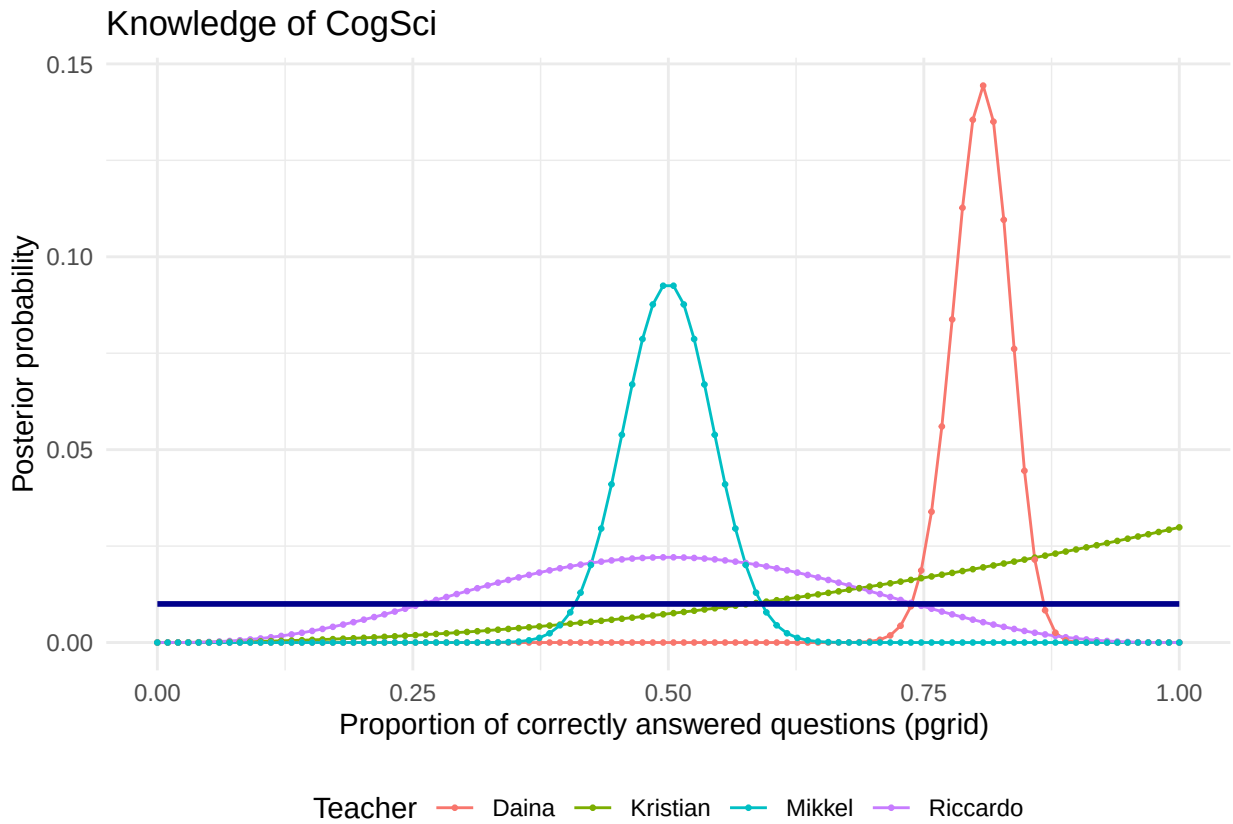
```r
# plotting the quadratic approximation
curve(dnorm(x, 0.5, 0.2), lty = 2)
```



- N.B. for the rest of the exercise just keep using the grid approximation (we'll move to quadratic approximations in two classes)

2. Estimate all the teachers' knowledge of CogSci. Who's best? Use grid approximation. Comment on the posteriors of Riccardo and Mikkel. 2a. Produce plots of the prior, and posterior for each teacher.

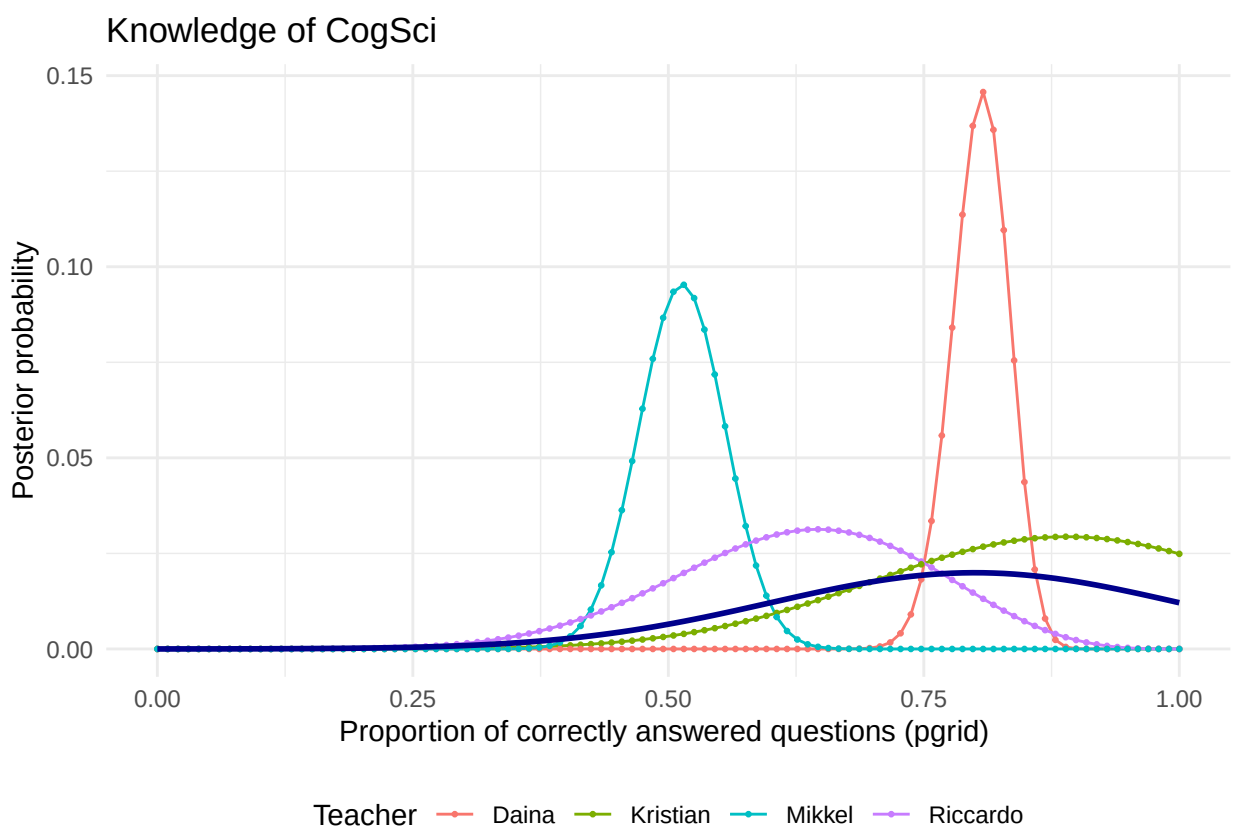
```r
############################ ALL THE TEACHERS KNOWLEDGE ################################
dens = 100

# Making the grid and prior
df <- tibble(
  p_grid = seq(0, 1, length.out = dens),
  prior = 1
) %>%
# Calculating likelihood and posterior
  mutate(
    likelihood_Riccardo = dbinom(3, size = 6, prob = p_grid),
    likelihood_Kristian = dbinom(2, size = 2, prob = p_grid),
    likelihood_Daina = dbinom(160, size = 198, prob = p_grid),
    likelihood_Mikkel = dbinom(66, size = 132, prob = p_grid),
    across(starts_with("likelihood"), ~ .x*prior), # unstd. posterior
    across(starts_with("likelihood"), ~ .x/sum(.x)) # std. posterior
  ) %>%
  pivot_longer(
    cols = starts_with("likelihood"),
    names_to = "Teacher",
    values_to = "Posterior"
  ) %>%
  separate(Teacher, c(NA, "Teacher"), sep = "_", convert = TRUE) # Removing likelihood from the factor

# PLOTTING
ggplot(df) +
  aes(p_grid, Posterior, colour = Teacher)+
  geom_line() +
  geom_point(size = 0.5) +
  geom_line(aes(p_grid, prior/dens), color = 'dark blue', size = 1) +  # Prior line
  labs(x = 'Proportion of correctly answered questions (pgrid) ',
       y = 'Posterior probability',
       title = 'Knowledge of CogSci') +
  theme_minimal()+
  theme(legend.position = 'bottom')
```



3. Change the prior. Given your teachers have all CogSci jobs, you should start with a higher appreciation of their knowledge: the prior is a normal distribution with a mean of 0.8 and a standard deviation of 0.2. Do the results change (and if so how)? 3a. Produce plots of the prior and posterior for each teacher.

```
############################ ALL TEACHERS WITH NORMAL PRIOR ################################
dens = 100

# Making the grid and prior
df_normal <- tibble(
  p_grid = seq(0, 1, length.out = dens),
  prior = dnorm(p_grid, mean = 0.8, sd = 0.2)
) %>%
# Calculating likelihood and posterior
  mutate(
    likelihood_Riccardo = dbinom(3, size = 6, prob = p_grid),
    likelihood_Kristian = dbinom(2, size = 2, prob = p_grid),
    likelihood_Daina = dbinom(160, size = 198, prob = p_grid),
    likelihood_Mikkel = dbinom(66, size = 132, prob = p_grid),
    across(starts_with("likelihood"), ~ .x*prior), # unstd. posterior
    across(starts_with("likelihood"), ~ .x/sum(.x)) # std. posterior
  ) %>%
  pivot_longer(
    cols = starts_with("likelihood"),
    names_to = "Teacher",
```

```
    values_to = "Posterior"
  )%>%
  separate(Teacher, c(NA, "Teacher"), sep = "_", convert = TRUE)


# PLOTTING
ggplot(df_normal) +
  aes(p_grid, Posterior, colour = Teacher)+
  geom_line() +
  geom_point(size = 0.5) +
  geom_line(aes(p_grid, prior/dens), color = 'dark blue', size = 1) +  # Prior line
  labs(x = 'Proportion of correctly answered questions (pgrid) ',
       y = 'Posterior probability',
       title = 'Knowledge of CogSci') +
  theme_minimal()+
  theme(legend.position = 'bottom')
```



How do the results change: - (ofc) prior is no longer flat - biggest difference in Riccardo and Kristian's posterior distribution (lower SD) - the posterior distr. of all the teachers have moved closer to 1. - the distributions with the least data have changed the most (Ric and Kris) - question: 'toppunkt' –> mere pointy, more narrow distribution - it is possible that all of the distributions have become more narrow, but the distributions with the least data are the ones where the change is most visible.

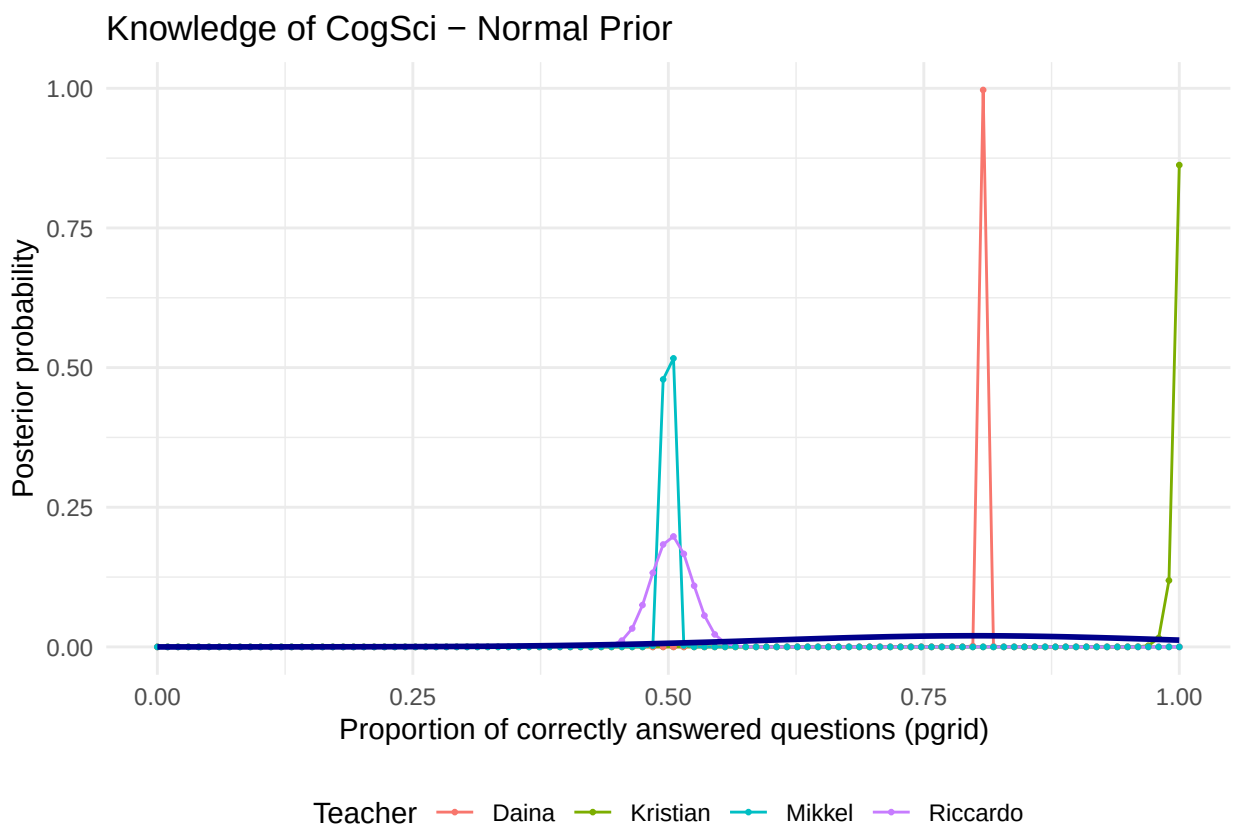
4. You go back to your teachers and collect more data (multiply the previous numbers by 100). Calculate their knowledge with both a uniform prior and a normal prior with a mean of 0.8 and a standard deviation of 0.2. Do you still see a difference between the results? Why?

```r
##############################  THE UNIFORM PRIOR * 100 OBS ########################################
dens = 100

# Making the grid and prior
df_uniform <- tibble(
  p_grid = seq(0, 1, length.out = dens),
  prior = 1
) %>%
# Calculating likelihood and posterior
  mutate(
    likelihood_Riccardo = dbinom(3*100, size = 6*100, prob = p_grid),
    likelihood_Kristian = dbinom(2*100, size = 2*100, prob = p_grid),
    likelihood_Daina = dbinom(160*100, size = 198*100, prob = p_grid),
    likelihood_Mikkel = dbinom(66*100, size = 132*100, prob = p_grid),
    across(starts_with("likelihood"), ~ .x*prior), # unstd. posterior
    across(starts_with("likelihood"), ~ .x/sum(.x)) # std. posterior
  ) %>%
  pivot_longer(
    cols = starts_with("likelihood"),
    names_to = "Teacher",
    values_to = "Posterior"
  )%>%
  separate(Teacher, c(NA, "Teacher"), sep = "_", convert = TRUE)
```

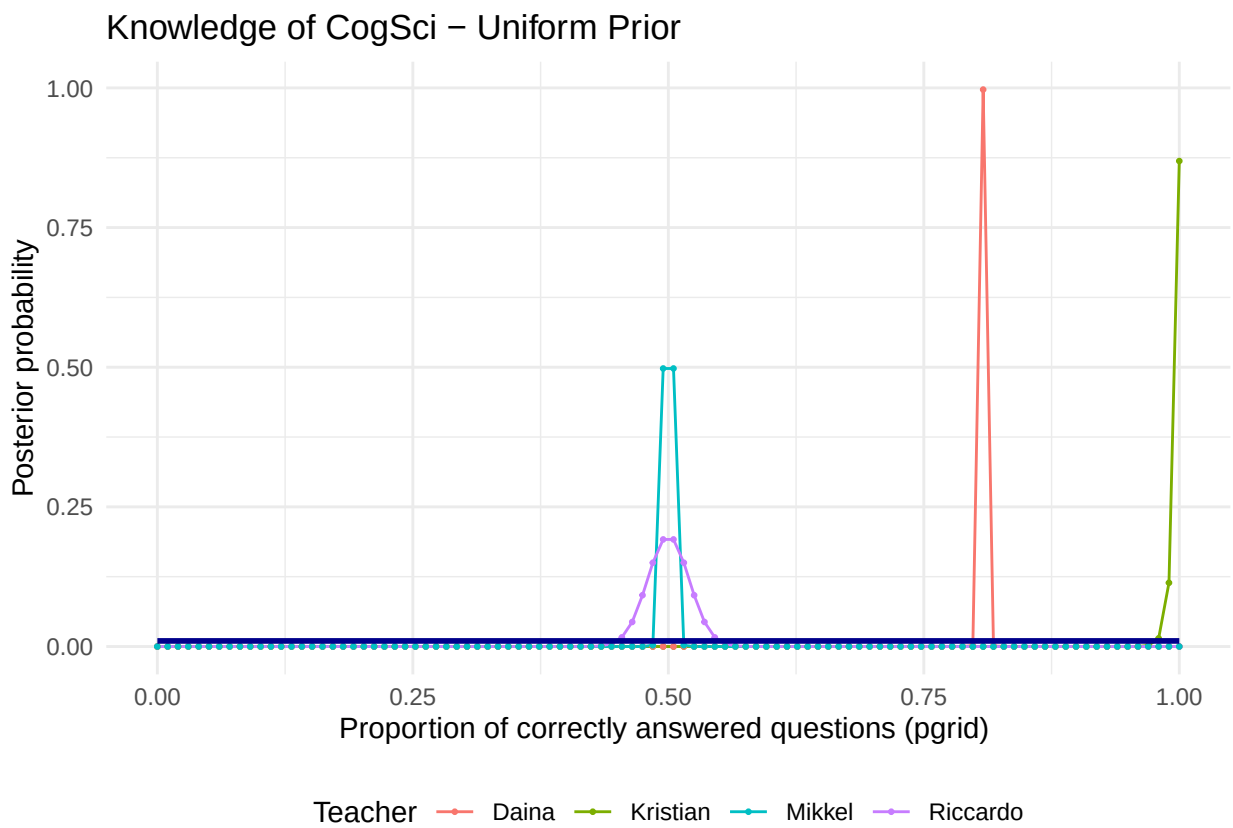
```r
##############################  THE NORMAL PRIOR * 100 OBS ################################
# Making the grid and prior
df_normal <- tibble(
  p_grid = seq(0, 1, length.out = dens),
  prior = dnorm(p_grid, mean = 0.8, sd = 0.2)
) %>%
# Calculating likelihood and posterior
  mutate(
    likelihood_Riccardo = dbinom(3*100, size = 6*100, prob = p_grid),
    likelihood_Kristian = dbinom(2*100, size = 2*100, prob = p_grid),
    likelihood_Daina = dbinom(160*100, size = 198*100, prob = p_grid),
    likelihood_Mikkel = dbinom(66*100, size = 132*100, prob = p_grid),
    across(starts_with("likelihood"), ~ .x*prior), # unstd. posterior
    across(starts_with("likelihood"), ~ .x/sum(.x)) # std. posterior
  ) %>%
  pivot_longer(
    cols = starts_with("likelihood"),
    names_to = "Teacher",
    values_to = "Posterior"
  )%>%
  separate(Teacher, c(NA, "Teacher"), sep = "_", convert = TRUE)
```

```r
############################ PLOTTING ########################################
# Uniform prior
plot_uniform <- ggplot(df_uniform) +
  aes(p_grid, Posterior, colour = Teacher)+
  geom_line() +
  geom_point(size = 0.5) +
  geom_line(aes(p_grid, prior/dens), color = 'dark blue', size = 1) +  # Prior line
```

```r
  labs(x = 'Proportion of correctly answered questions (pgrid) ',
       y = 'Posterior probability',
       title = 'Knowledge of CogSci - Uniform Prior') +
  theme_minimal()+
  theme(legend.position = 'bottom')



# Normal prior
plot_normal <- ggplot(df_normal) +
  aes(p_grid, Posterior, colour = Teacher)+
  geom_line() +
  geom_point(size = 0.5) +
  geom_line(aes(p_grid, prior/dens), color = 'dark blue', size = 1) +  # Prior line
  labs(x = 'Proportion of correctly answered questions (pgrid) ',
       y = 'Posterior probability',
       title = 'Knowledge of CogSci - Normal Prior') +
  theme_minimal()+
  theme(legend.position = 'bottom')

plot_normal
```




```r
plot_uniform
```

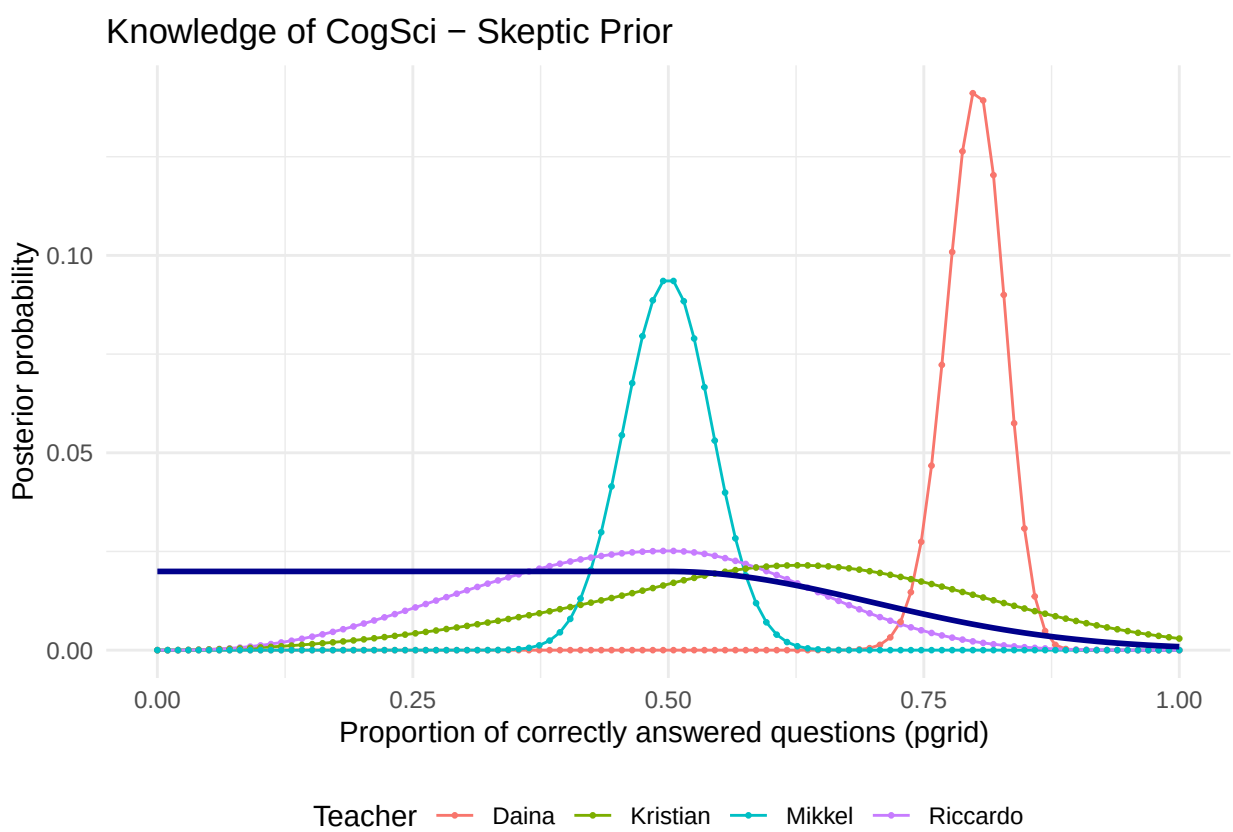


As the amount of data increase, the 'tops' become higher and more pointy. That is, the model is more certain that a teacher will have a given parameter distribution. More narrow distributions. The prior does not have a great effect on the differents distributions, but there is a small difference between using the uniform flat prior and the normally distributed prior.

5. Imagine you're a skeptic and think your teachers do not know anything about CogSci, given the content of their classes. How would you operationalize that belief?

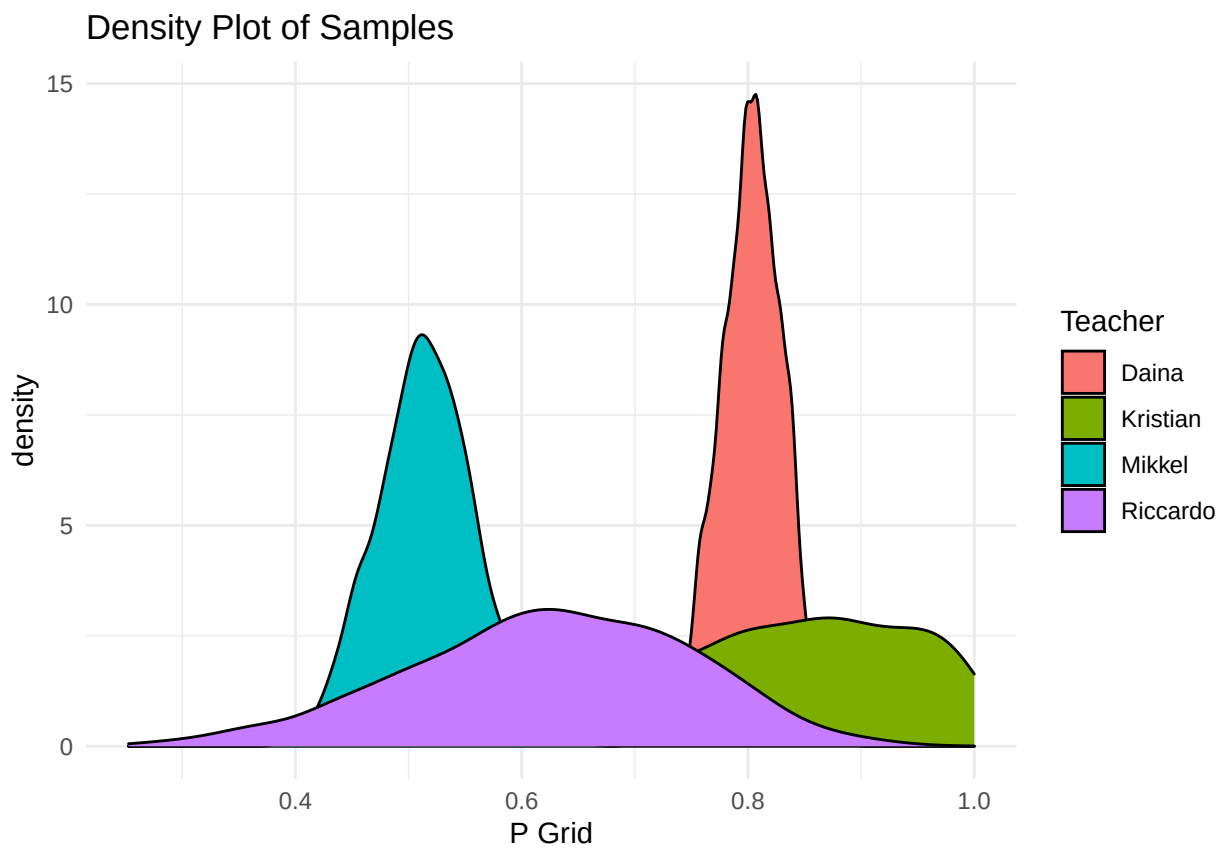
```
##############################  THE SKEPTIC PRIOR ##############################################
dens = 100

# Making the grid and prior
df_skeptic <- tibble(
  p_grid = seq(0, 1, length.out = dens),
  prior = ifelse(p_grid <= 0.5,
                 max(dnorm(p_grid, mean = 0.5, sd = 0.2)),
                 dnorm(p_grid, mean = 0.5, sd = 0.2))
) %>%
# Calculating likelihood and posterior
  mutate(
    likelihood_Riccardo = dbinom(3, size = 6, prob = p_grid),
    likelihood_Kristian = dbinom(2, size = 2, prob = p_grid),
    likelihood_Daina = dbinom(160, size = 198, prob = p_grid),
    likelihood_Mikkel = dbinom(66, size = 132, prob = p_grid),
    across(starts_with("likelihood"), ~ .x*prior), # unstd. posterior
    across(starts_with("likelihood"), ~ .x/sum(.x)) # std. posterior
```

```r
  ) %>%
  pivot_longer(
    cols = starts_with("likelihood"),
    names_to = "Teacher",
    values_to = "Posterior"
  )%>%
  separate(Teacher, c(NA, "Teacher"), sep = "_", convert = TRUE)

# PLOTTING
ggplot(df_skeptic) +
  aes(p_grid, Posterior, colour = Teacher)+
  geom_line() +
  geom_point(size = 0.5) +
  geom_line(aes(p_grid, prior/dens), color = 'dark blue', size = 1) +  # Prior line
  labs(x = 'Proportion of correctly answered questions (pgrid) ',
       y = 'Posterior probability',
       title = 'Knowledge of CogSci - Skeptic Prior') +
  theme_minimal()+
  theme(legend.position = 'bottom')
```



6. Optional question: Can you estimate the difference between Riccardo's estimated knowledge and that of each of the other teachers? Would you deem it credible (that is, would you believe that it is actually different)?

```r
######################## DIFFERENCE BETWEEN RICCARDO AND OTHER TEACHERS ###################

# Setting seed so the sampling can be replicated
set.seed(123)

# Defining the distribution for each teacher
dens = 100

# Making the grid and prior
df <- tibble(
  p_grid = seq(0, 1, length.out = dens),
  prior = dnorm(p_grid, mean = 0.8, sd = 0.2)
) %>%
# Calculating likelihood and posterior
  mutate(
    likelihood_Riccardo = dbinom(3, size = 6, prob = p_grid),
    likelihood_Kristian = dbinom(2, size = 2, prob = p_grid),
    likelihood_Daina = dbinom(160, size = 198, prob = p_grid),
    likelihood_Mikkel = dbinom(66, size = 132, prob = p_grid),
    across(starts_with("likelihood"), ~ .x*prior), # unstd. posterior
    across(starts_with("likelihood"), ~ .x/sum(.x)) # std. posterior
  )

# Sampling from each teachers distribution
sampling_size = 1000
samples <- tibble(
  Index = seq(1, sampling_size),
  Riccardo = sample(df$p_grid, size = sampling_size, prob = df$likelihood_Riccardo ,replace = T),
  Kristian = sample(df$p_grid, size = sampling_size, prob = df$likelihood_Kristian ,replace = T),
  Daina = sample(df$p_grid, size = sampling_size, prob = df$likelihood_Daina ,replace = T),
  Mikkel = sample(df$p_grid, size = sampling_size, prob = df$likelihood_Mikkel ,replace = T)
)  %>%
  pivot_longer(cols = -c(Index),
               names_to = "Teacher",
               values_to = "Samples")

# Plotting
ggplot(samples)+
  aes(x = Samples, fill = Teacher)+
  geom_density()+
  labs(x = "P Grid", title = "Density Plot of Samples")+
  theme_minimal()
```




```r
# Releveling the factor and doing a lm
samples <- samples %>%
  mutate(Teacher = as.factor(Teacher),
         Teacher = fct_relevel(Teacher, "Riccardo"))

# lm
lm(Samples ~ Teacher, data = samples) %>%
  summary
```

```
##
## Call:
## lm(formula = Samples ~ Teacher, data = samples)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.52618 -0.03897  0.00143  0.04184  0.33477
##
## Coefficients:
##                  Estimate Std. Error t value Pr(>|t|)
## (Intercept)      0.624828   0.002999  208.32   <2e-16 ***
## TeacherDaina     0.178556   0.004242   42.09   <2e-16 ***
## TeacherKristian  0.184182   0.004242   43.42   <2e-16 ***
## TeacherMikkel   -0.110566   0.004242  -26.07   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
```

```
## Residual standard error: 0.09485 on 3996 degrees of freedom
## Multiple R-squared:  0.6335, Adjusted R-squared:  0.6332
## F-statistic:  2302 on 3 and 3996 DF,  p-value: < 2.2e-16
```

Second part: Focusing on predictions

Last year you assessed the teachers (darned time runs quick!). Now you want to re-test them and assess whether your models are producing reliable predictions. In Methods 3 we learned how to do machine-learning style assessment of predictions (e.g. rmse on testing datasets). Bayesian stats makes things a bit more complicated. So we'll try out how that works. N.B. You can choose which prior to use for the analysis of last year's data.

Questions to be answered (but see guidance below): 1- Write a paragraph discussing how assessment of prediction performance is different in Bayesian vs. frequentist models 2- Provide at least one plot and one written line discussing prediction errors for each of the teachers.

This is the old data: - Riccardo: 3 correct answers out of 6 questions - Kristian: 2 correct answers out of 2 questions (then he gets bored) - Daina: 160 correct answers out of 198 questions (Daina never gets bored) - Mikkel: 66 correct answers out of 132 questions

This is the new data: - Riccardo: 9 correct answers out of 10 questions (then he freaks out about teaching preparation and leaves) - Kristian: 8 correct answers out of 12 questions - Daina: 148 correct answers out of 172 questions (again, Daina never gets bored) - Mikkel: 34 correct answers out of 65 questions

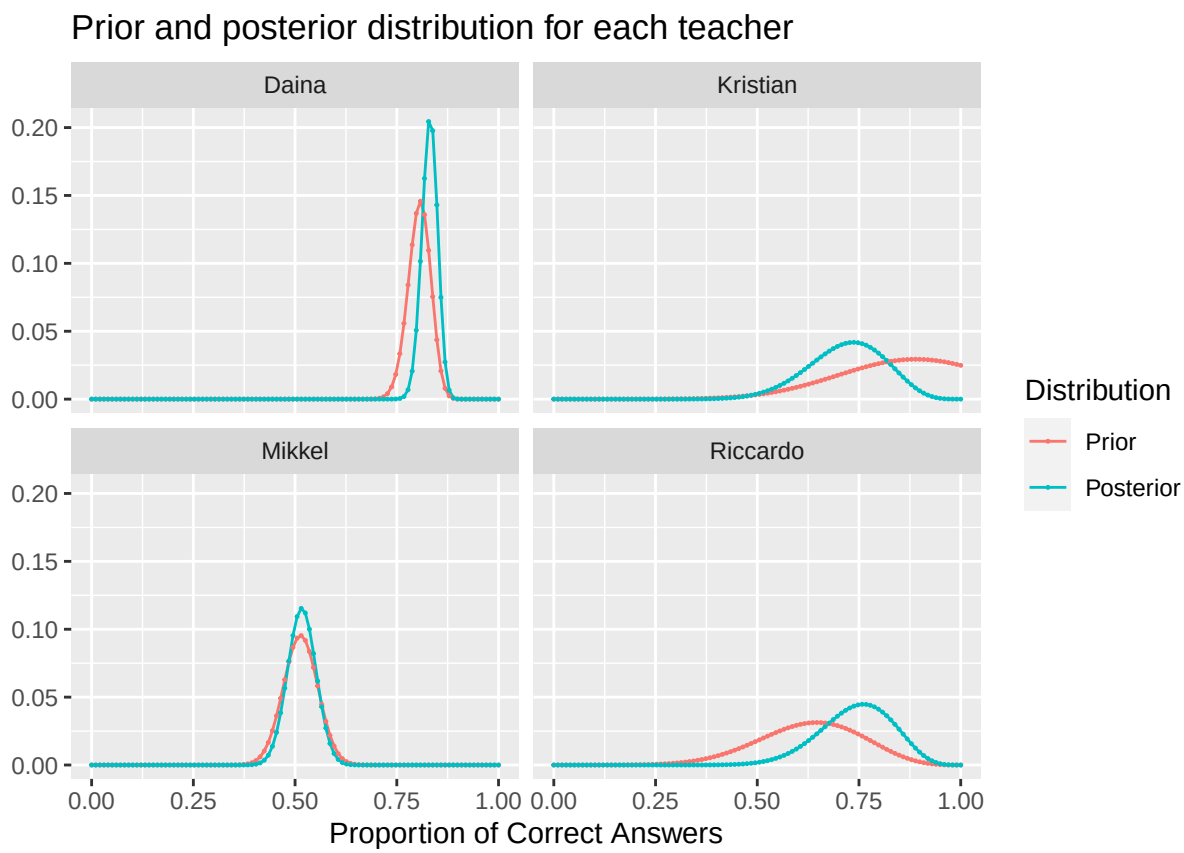
Guidance Tips

1. There are at least two ways of assessing predictions.
2. Last year's results are this year's expectations.
3. Are the parameter estimates changing? (way 1)

```r
###################################### WAY 1 ##############################################
########### Creating posterior dist. with prior from the old posterior distr. #############

# Length of grid
sequence_length = 100

# Making the grid and priors of last years posterior distributions.
df_new <- tibble(
  p_grid = seq(0, 1, length.out = sequence_length),
  prior_Riccardo = df$likelihood_Riccardo,
  prior_Kristian = df$likelihood_Kristian,
  prior_Daina = df$likelihood_Daina,
  prior_Mikkel = df$likelihood_Mikkel
) %>%
# Calculating likelihood and posterior
  mutate(
    likelihood_Riccardo = dbinom(9, size = 10, prob = p_grid),
    likelihood_Kristian = dbinom(8, size = 12, prob = p_grid),
    likelihood_Daina = dbinom(148, size = 172, prob = p_grid),
    likelihood_Mikkel = dbinom(34, size = 65, prob = p_grid),
# Standardizing
    posterior_Riccardo = (likelihood_Riccardo*prior_Riccardo)/sum(likelihood_Riccardo*prior_Riccardo),
    posterior_Kristian = (likelihood_Kristian*prior_Kristian)/sum(likelihood_Kristian*prior_Kristian),
    posterior_Daina = (likelihood_Daina*prior_Daina)/sum(likelihood_Daina*prior_Daina),
    posterior_Mikkel = (likelihood_Mikkel*prior_Mikkel)/sum(likelihood_Mikkel*prior_Mikkel)
```

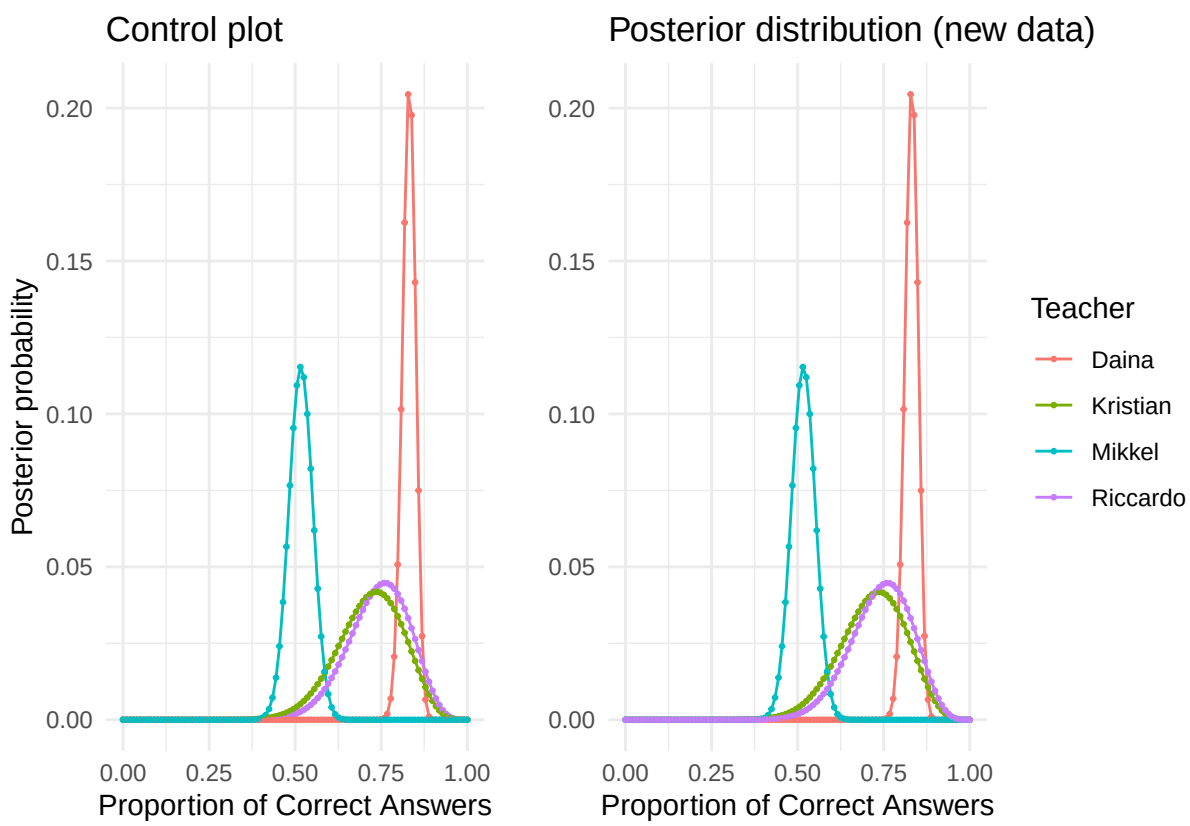
```r
  ) %>%
  select(
    -c('likelihood_Riccardo', 'likelihood_Kristian', 'likelihood_Daina', 'likelihood_Mikkel')
    )

# Making a new data_frame with only priors
df_prior <- df_new %>%
  select(p_grid,starts_with("prior")) %>%
  pivot_longer(
    cols = starts_with("prior"),
    names_to = "Teacher",
    values_to = "Prior"
  )%>%
  separate(Teacher, c(NA, "Teacher"), sep = "_", convert = TRUE)

# Making a new data_frame with only posteriors
df_post <- df_new %>%
  select(p_grid, starts_with("posterior")) %>%
  pivot_longer(
    cols = starts_with("posterior"),
    names_to = "Teacher",
    values_to = "Posterior"
  )%>%
  separate(Teacher, c(NA, "Teacher"), sep = "_", convert = TRUE)

# Merging
df_1 <- merge(df_post, df_prior)
```

```r
########## PLOT 1

df_1 %>% pivot_longer(
  cols = c('Prior', 'Posterior'),
  names_to = 'Distribution',
  values_to = 'value') %>%
  mutate(
    Distribution = relevel(as.factor(Distribution), 'Prior')
  ) %>%
  ggplot(aes(p_grid, value, colour = Distribution))+
  geom_line()+
  geom_point(size = 0.2) +
  facet_wrap(~Teacher) +
  labs(x = "Proportion of Correct Answers", y = "", title = "Prior and posterior distribution for each t
```




```
######################### Control plot ###################
# Checking whether there is a difference between the new posterior distribution and the post. dist. of

dens = 100

# Making the grid and prior
df_control <- tibble(
  p_grid = seq(0, 1, length.out = dens),
  prior = dnorm(p_grid, mean = 0.8, sd = 0.2)
) %>%
# Calculating likelihood and posterior
  mutate(
    likelihood_Riccardo = dbinom(12, size = 16, prob = p_grid),
    likelihood_Kristian = dbinom(10, size = 14, prob = p_grid),
    likelihood_Daina = dbinom(308, size = 370, prob = p_grid),
    likelihood_Mikkel = dbinom(100, size = 197, prob = p_grid),
    across(starts_with("likelihood"), ~ .x*prior), # unstd. posterior
    across(starts_with("likelihood"), ~ .x/sum(.x)) # std. posterior
  ) %>%
  pivot_longer(
    cols = starts_with("likelihood"),
    names_to = "Teacher",
    values_to = "Posterior"
  )%>%
  separate(Teacher, c(NA, "Teacher"), sep = "_", convert = TRUE)
```

```r
# PLOTTING
plot_control <- ggplot(df_control) +
  aes(p_grid, Posterior, colour = Teacher)+
  geom_line() +
  geom_point(size = 0.5) +
  #geom_line(aes(p_grid, prior/dens), color = 'dark blue', size = 1) +  # Prior line
  labs(x = 'Proportion of Correct Answers',
       y = 'Posterior probability',
       title = 'Control plot') +
  theme_minimal()+
  theme(legend.position = 'none')

plot_post <-
  ggplot(df_1, aes(p_grid, Posterior, color = Teacher))+
  geom_line()+
  geom_point(size = 0.5)+
  labs(x = "Proportion of Correct Answers", y = "", title = 'Posterior distribution (new data)') +
  theme_minimal()

plot_control + plot_post
```

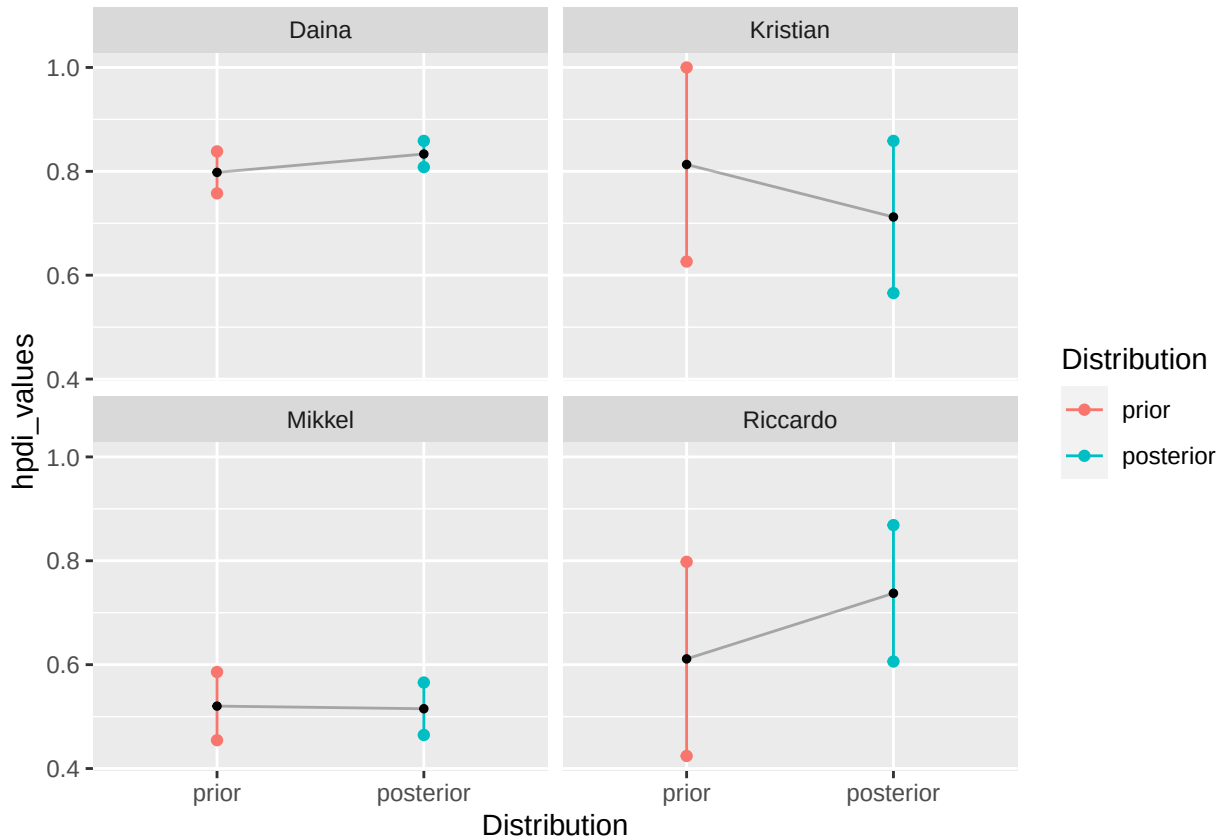



```r
##################### COMPARING DISTRIBUTION PARAMETERS  #######################
# first - sampling the posterior distributions of the old and the new data, respectively

# sample_size
```

```r
sample_size = 1000

# Creating a data_frame with samples from all the teachers
samples <- tibble(
  # Samples from prior distribution
  prior_Riccardo = sample(df$p_grid, prob = df_new$prior_Riccardo, size = sample_size, replace = T),
  prior_Kristian = sample(df$p_grid, prob = df_new$prior_Kristian, size = sample_size, replace = T),
  prior_Daina = sample(df$p_grid, prob = df_new$prior_Daina, size = sample_size, replace = T),
  prior_Mikkel = sample(df$p_grid, prob = df_new$prior_Mikkel, size = sample_size, replace = T),

  # Samples from the posterior distribution
  posterior_Riccardo = sample(df$p_grid, prob = df_new$posterior_Riccardo, size = sample_size, replace =
  posterior_Kristian = sample(df$p_grid, prob = df_new$posterior_Kristian, size = sample_size, replace =
  posterior_Daina = sample(df$p_grid, prob = df_new$posterior_Daina, size = sample_size, replace = T),
  posterior_Mikkel = sample(df$p_grid, prob = df_new$posterior_Mikkel, size = sample_size, replace = T)
  )

# Getting HPDI-values
hpdi <- tibble(
  name = colnames(samples),
  hpdi_values = lapply(samples, HPDI)) %>%
  unnest(hpdi_values) %>%
  separate(name, c("Distribution","Teacher"), sep = "_", convert = T) %>%
  mutate(
    Distribution = relevel(as.factor(Distribution), 'prior')
  )
```

```r
# PLOTTING plot 2 ?
hpdi %>%
  ggplot(aes(x = Distribution, y = hpdi_values, color = Distribution))+
  geom_point()+
  geom_line()+
  stat_summary(fun = "mean", geom = "point", color = "black", size = 1)+
  stat_summary(fun.y = "mean", geom = "line", aes(group = "prior"), color = "black", alpha = 0.3)+
  facet_wrap(~Teacher)
```

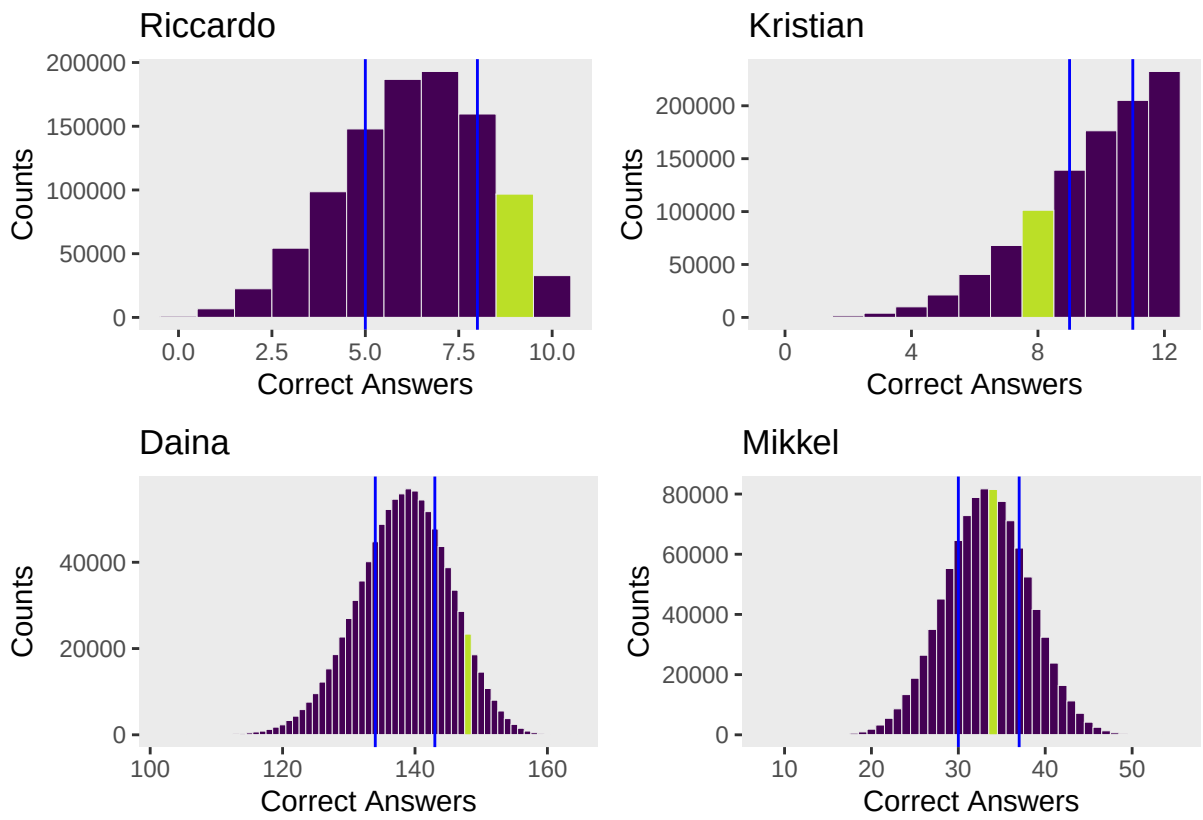
```
## Warning: `fun.y` is deprecated. Use `fun` instead.
```



4. How does the new data look in last year's predictive posterior? (way 2)

```
#################### PREDICTING NEW DATA FROM OLD POSTERIOR ########################

# Creating new samples from the posteriors of the old data.
sample_size = 1000

samples_old <- tibble(
  Riccardo = sample(df_new$p_grid, prob = df_new$prior_Riccardo, size = sample_size, replace = T),
  Kristian = sample(df_new$p_grid, prob = df_new$prior_Kristian, size = sample_size, replace = T),
  Daina = sample(df_new$p_grid, prob = df_new$prior_Daina, size = sample_size, replace = T),
  Mikkel = sample(df_new$p_grid, prob = df_new$prior_Mikkel, size = sample_size, replace = T))

# With the samples from the posterior distribution of the old data, we can create simulations of how th
simulation_size = 1e6

simulations <- tibble(
  Riccardo = rbinom(simulation_size, size = 10, prob = samples_old$Riccardo),
  Kristian = rbinom(simulation_size, size = 12, prob = samples_old$Kristian),
  Daina = rbinom(simulation_size, size = 172, prob = samples_old$Daina),
  Mikkel = rbinom(simulation_size, size = 65, prob = samples_old$Mikkel)
  ) %>%
  pivot_longer(
    cols = everything(),
    names_to = "Teacher",
```

```
    values_to = "correct_answers"
  )
```

```
############ PLOTTING ############
# Riccardo
ric_plot <- simulations %>%
  filter(Teacher == "Riccardo") %>%
  ggplot(aes(x = correct_answers))+
  geom_histogram(aes(fill = correct_answers == 9),
                 binwidth = 1, center = 0,
                 color = "grey92", size = 1/10)+
  scale_fill_viridis_d(option = "D", end = .9) +
  geom_vline(xintercept = PI(simulations$correct_answers[simulations$Teacher == "Riccardo"], prob = 0.5
  geom_vline(xintercept = PI(simulations$correct_answers[simulations$Teacher == "Riccardo"], prob = 0.5
  labs(x = "Correct Answers", y = "Counts", title = "Riccardo") +
  theme(legend.position = "none",
        panel.grid = element_blank())

# Kristian
kri_plot <- simulations %>%
  filter(Teacher == "Kristian") %>%
  ggplot(aes(x = correct_answers))+
  geom_histogram(aes(fill = correct_answers == 8),
                 binwidth = 1, center = 0,
                 color = "grey92", size = 1/10)+
  scale_fill_viridis_d(option = "D", end = .9) +
  geom_vline(xintercept = PI(simulations$correct_answers[simulations$Teacher == "Kristian"], prob = 0.5
  geom_vline(xintercept = PI(simulations$correct_answers[simulations$Teacher == "Kristian"], prob = 0.5
  labs(x = "Correct Answers", y = "Counts", title = "Kristian") +
  theme(legend.position = "none",
        panel.grid = element_blank())

# Daina
dai_plot <- simulations %>%
  filter(Teacher == "Daina") %>%
  ggplot(aes(x = correct_answers))+
  geom_histogram(aes(fill = correct_answers == 148),
                 binwidth = 1, center = 0,
                 color = "grey92", size = 1/10)+
  scale_fill_viridis_d(option = "D", end = .9) +
  geom_vline(xintercept = PI(simulations$correct_answers[simulations$Teacher == "Daina"], prob = 0.50)[
  geom_vline(xintercept = PI(simulations$correct_answers[simulations$Teacher == "Daina"], prob = 0.50)[
  labs(x = "Correct Answers", y = "Counts", title = "Daina") +
  theme(legend.position = "none",
        panel.grid = element_blank())

# Mikkel
mik_plot <- simulations %>%
  filter(Teacher == "Mikkel") %>%
  ggplot(aes(x = correct_answers))+
  geom_histogram(aes(fill = correct_answers == 34),
                 binwidth = 1, center = 0,
                 color = "grey92", size = 1/10)+
```

```
  scale_fill_viridis_d(option = "D", end = .9) +
  geom_vline(xintercept = PI(simulations$correct_answers[simulations$Teacher == "Mikkel"], prob = 0.50)
  geom_vline(xintercept = PI(simulations$correct_answers[simulations$Teacher == "Mikkel"], prob = 0.50)
  labs(x = "Correct Answers", y = "Counts", title = "Mikkel") +
  theme(legend.position = "none",
        panel.grid = element_blank())


(ric_plot + kri_plot) / (dai_plot + mik_plot)
```